# midterm review

Chapters 1-6, 8

## Chapter 1: Introduction

→ why experiments?

→ why design?

- EU, Meas. Unit, placebo, control, etc

## Chapter 2: Randomization.

- Randomization to improve design control confounding.
- gives us a way to do inference.
- $\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2)$

## Chapter 3:

- Completely Randomized Design
- analysis via ANOVA.

- models, parameters, ANOVA tables.
- $H_0: \mu_i = \mu_j$ for all $i, j$

# Chapter 4: Contrasts.

- "Post-Hoc" analysis
- Investigate ANOVA results further.

$(1,-1,0)$

$\mu_1-\mu_2 \quad \mu_2-\mu_3$

$\mu_1-\mu_3$

] pairwise comparison

pairwise contrasts.

if 1 is control,

$\mu_1 - \left(\frac{\mu_2+\mu_3}{2}\right) \overset{?}{=} 0$ is control different from other 2 groups?

$(1, -\frac{1}{2}, -\frac{1}{2})$

**Contrast:** $W = w_1\mu_1 + w_2\mu_2 + \ldots + w_g\mu_g$

$$\sum_{i=1}^{g} w_i = 0$$

$-2\mu_1 + \mu_2 + \mu_3$

$W^{(1)} = (1, -\frac{1}{2}, -\frac{1}{2})$ $\quad W^{(2)} = (-2, 1, 1)$

$W^{(k)} = \beta W^{(l)}$, then you get same p-value.

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on paper:

$(w_1, w_2, \ldots, w_g) \rightarrow (\mu_1, \mu_2, \ldots, \mu_g)$

$\mu_5 - \mu_3 \rightarrow (0, 0, -1, 0, 1, \ldots, 0)$

$$\frac{\mu_4 + \mu_5}{2} - \frac{\mu_1 + \mu_3 + \mu_6}{3}$$

$$= \left(-\frac{1}{3}, 0, \frac{-1}{3}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{3}\right)$$

1 2 3 4

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y ~ factor(trt)  "water", "acid", "milk"

$\mu_1$ = mean of acid.

$\mu_2$ = mean of milk

$\mu_3$ = mean of water

"0"
"02"
"3"

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mod ← aov( y ~ factor(trt) , data = ... )

summary(mod)

library("emmeans")

emmeans( mod , ~ trt )

| | μ | SE |
|---|---|---|
| → acid | — | — |
| → milk | — | — |
| → water | — | — |

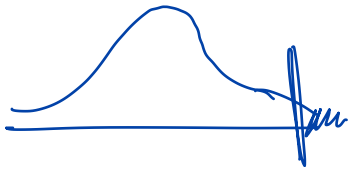
→ $(1, -\frac{1}{2}, -\frac{1}{2})$ = $\mu_{acid} - \frac{\mu_{milk} + \mu_{water}}{2}$

$$\hat{\psi} = \omega_1 \hat{\mu}_1 + \omega_2 \hat{\mu}_2 + \omega_3 \hat{\mu}_3$$

$$\sum \omega = 0$$

$\mu_1 - \mu_3 \overset{?}{=} 0$ ← $(1, 0, -1)$ → $w(\{\bar{y}_{i\cdot}\})$

$$t = \frac{(\hat{\mu}_1 - \hat{\mu}_3) - 0}{\sqrt{MS_E}\left(\sqrt{\sum_{i=1}^{3} \frac{w_i^2}{n_i}}\right)}$$



$t = 3.5$

$(1, 0, -1)$

$1 \cdot \bar{y}_{1\cdot} + 0 \cdot \bar{y}_{2\cdot} + (-1)\bar{y}_{3\cdot} = \hat{\mu}_1 - \hat{\mu}_3$

$P < 0.05$

reject.

conclude: $\mu_1 - \mu_3 \neq 0$,

so $\mu_1 \neq \mu_3$.

# Chapter 5: multiple comparisons

$\alpha = 0.05$ control Type I error rate to be at most 0.05

12 independent test, your prob of making at least one type I error is bigger than 50%.

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Controlling Family of errors.

→ per-comparison error rate.

→ Family wise error rate.

→ Strong FWER

→ False Discovery Rate ←

"FDR" → Benjamini-Hochberg.

→ Strong FWER

* Bonferroni
* Holm's

power(Holm's) ≥ power(Bonf)

Contrast Specific:

* Tukey's (HSD): all pairwise comparisons

$\mu_1-\mu_2,\ \mu_1-\mu_3,\ \mu_2-\mu_3$

* Dunnett: not all comparisons, but comparing to control

$\mu_2-\mu_1,\ \mu_3-\mu_1$

* Scheffé's: Any number of contrasts.

Controls data snooping.

least powerful

(if # test < 10)

# Chapter 6 :- 8

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

$$\varepsilon_{ijk} \overset{iid}{\sim} N(0, \sigma^2)$$

assumptions :

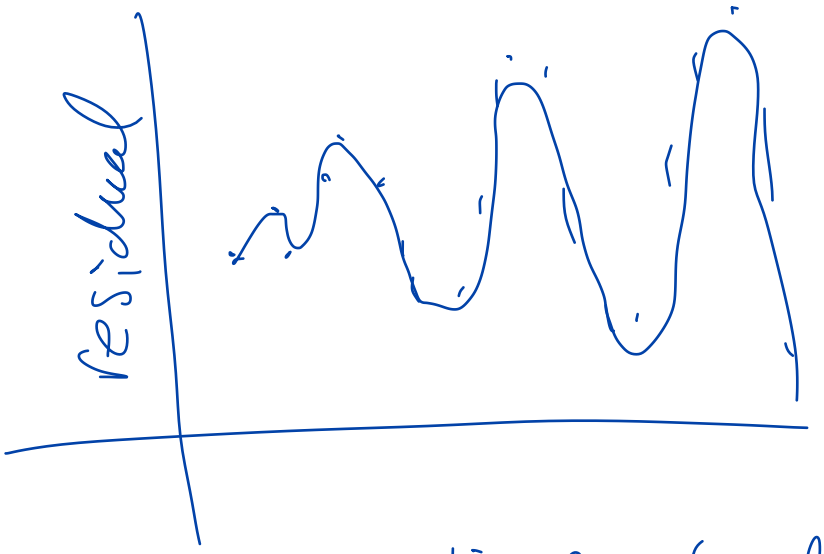
- ✓ Independence of $\varepsilon_{ijk}$
- Constant variance ($\sigma^2$) (homo-scedasticity)
- Normality.

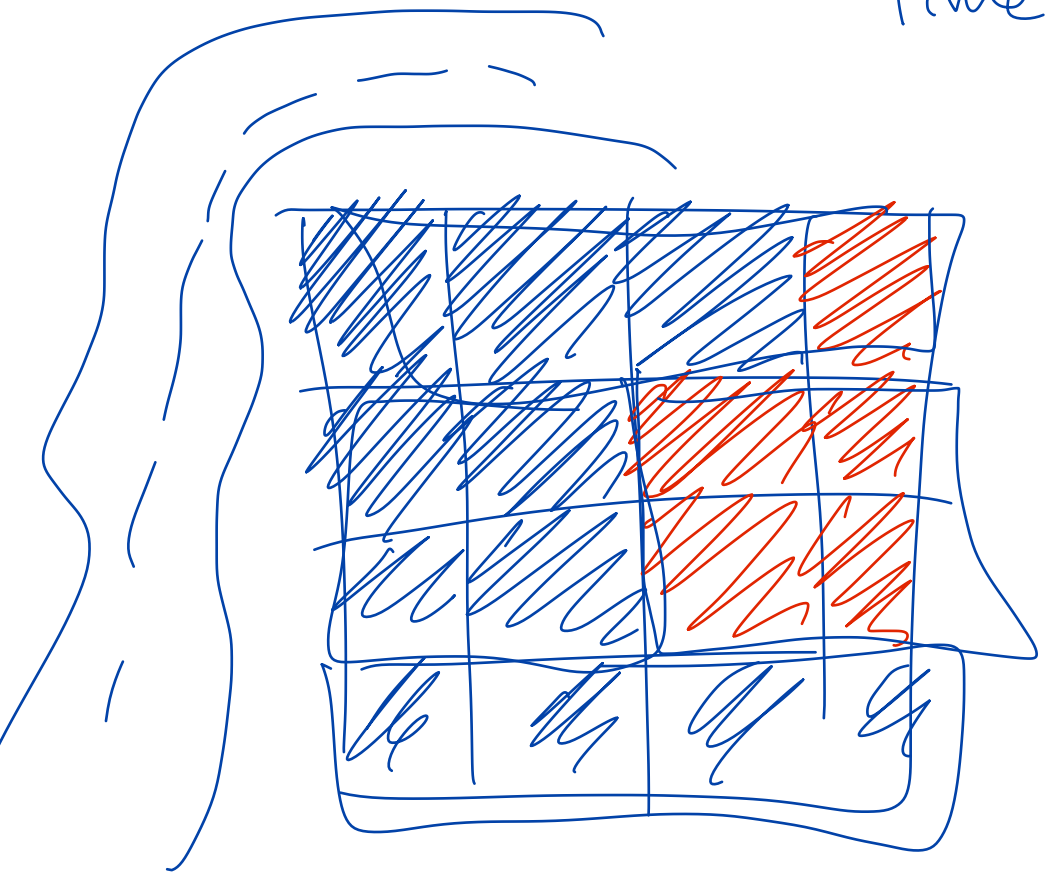
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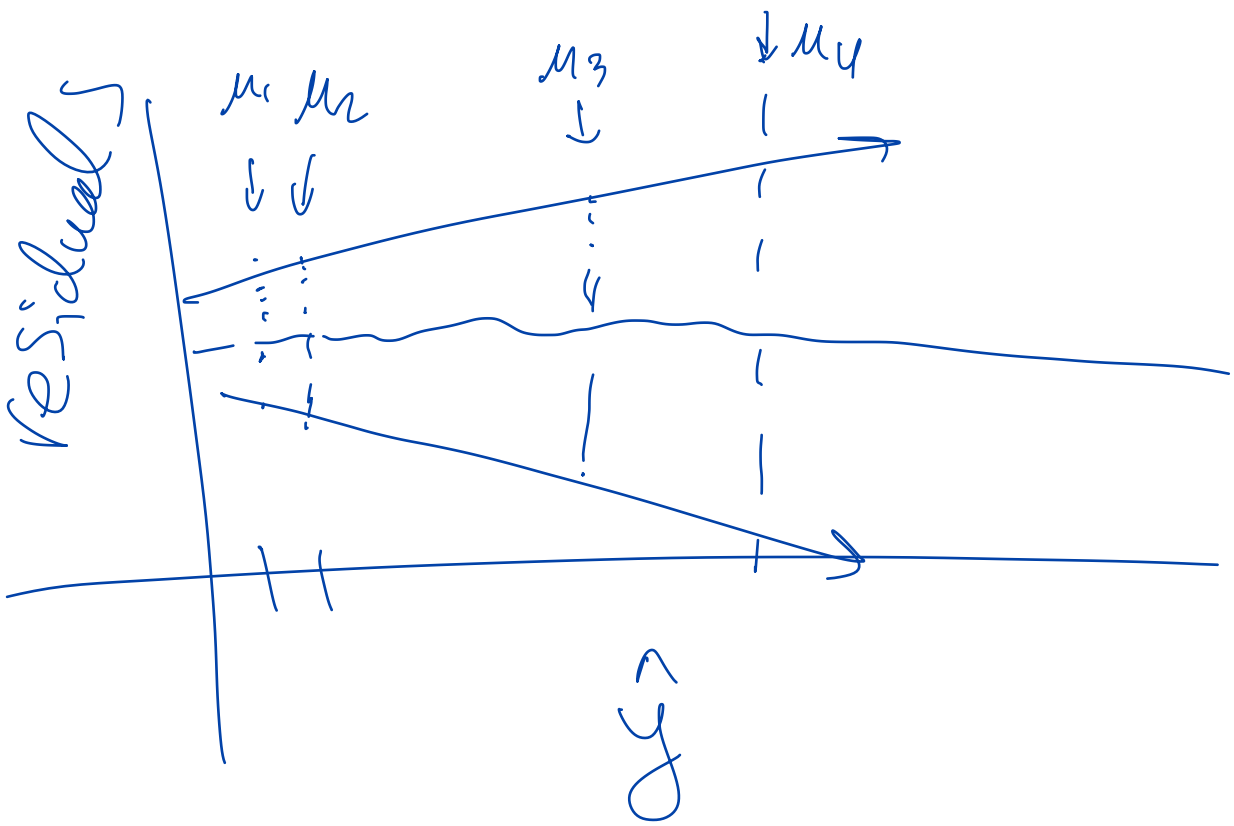
check for independence:

plot residuals over time, space, maybe group?

avoid:



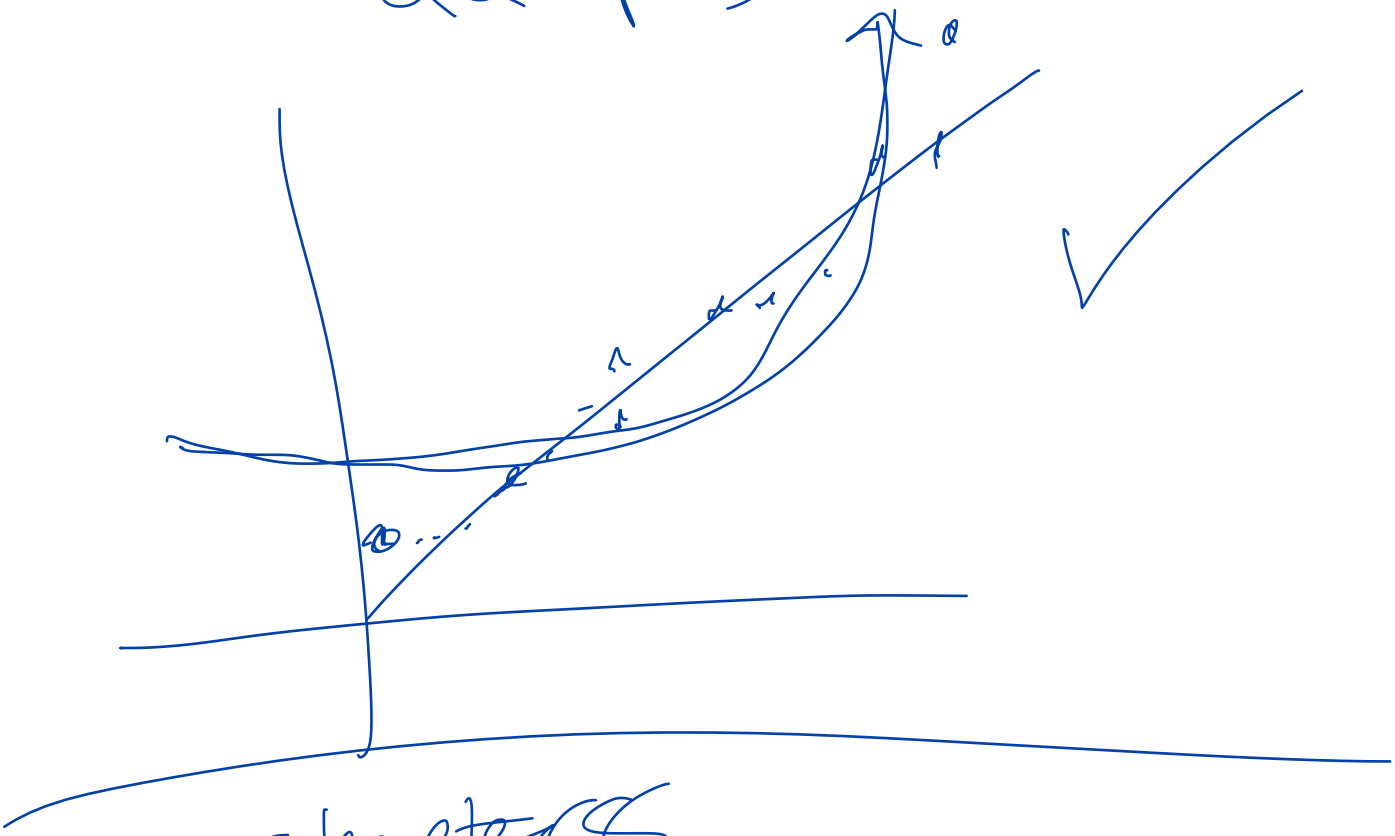






$y^{1/2}$, $y^{1/3}$, $\log(y)$

$y^p$, $p < 1$

QQ-plots



chapter 8

factorial design / analysis